Marlos C. Machado
CMPUT 365: Introduction to Reinforcement Learning

## Worksheet 3: Value Functions & Bellman Equations

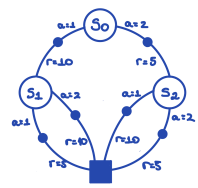
**Question 1.** In this question, you will take a word specification of an MDP, and write the formal terms and determine the optimal policy. Suppose you have a problem with two actions. The agent always starts in the same state, $s_0$. From this state, if it takes action 1 it transitions to a new state $s_1$ and receives reward 10; if it takes action 2 it transitions to a new state $s_2$ and receives reward 5. From $s_1$ if it takes action 1 it receives a reward of 5 and terminates; if it takes action 2 it receives a reward of 10 and terminates. From $s_2$ if it takes action 1 it receives a reward of 10 and terminates; if it takes action 2 it receives a reward of 5 and terminates. Assume the agent cares equally about long term reward as about immediate reward.

(a) Draw the MDP for this problem. Is it an episodic or continuing problem? What is $\gamma$?

(b) Assume the policy is $\pi(a = 1|s_i) = 0.3$ for all $s_i \in \{s_0, s_1, s_2\}$. What is $\pi(a = 2|s_i)$? And what is the value function for this policy? In other words, find $v_\pi(s)$ for all three states.

(c) What is the optimal policy in this environment?

(a) Episodic, thus $\gamma = 1$.




(b) $\pi(a = 2|s_i) = 1 - \pi(a = 1|s_i)\ \forall s_i \in \{s_0, s_1, s_2\}$
$= 1 - 0.3 = 0.7$

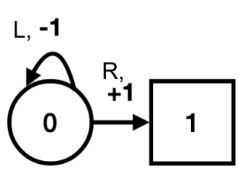
$v_\pi(s) = \sum_{a\in\mathcal{A}} \pi(a|s) \sum_{s'\in\mathcal{S}} \sum_{r\in\mathcal{R}} p(s', r|s, a)(r + \gamma v)_\pi(s'))\ \forall s \in \{s_0, s_1, s_2\}$

We can write $v_\pi(s)$ for state $s_0, s_1$, and $s_2$:

$v_\pi(s_0) = \sum_{a\in\mathcal{A}} \pi(a|s_0) \sum_{s'\in\mathcal{S}} \sum_{r\in\mathcal{R}} p(s', r|s_0, a)(r + \gamma v_\pi(s')) = 0.3 \cdot (10 + \gamma v_\pi(s_1)) + 0.7 \cdot (5 + \gamma v_\pi(s_2))$
$v_\pi(s_1) = \sum_{a\in\mathcal{A}} \pi(a|s_1) \sum_{s'\in\mathcal{S}} \sum_{r\in\mathcal{R}} p(s', r|s_1, a)(r + \gamma v_\pi(s')) = 0.3 \cdot (5 + \gamma \cdot 0) + 0.7 \cdot (10 + \gamma \cdot 0)$
$v_\pi(s_2) = \sum_{a\in\mathcal{A}} \pi(a|s_2) \sum_{s'\in\mathcal{S}} \sum_{r\in\mathcal{R}} p(s', r|s_2, a)(r + \gamma v_\pi(s')) = 0.3 \cdot (10 + \gamma \cdot 0) + 0.7 \cdot (5 + \gamma \cdot 0)$

(c) $v_\pi(s_2) = 3 + 3.5 = 6.5$
$v_\pi(s_1) = 1.5 + 7 = 8.5$
$v_\pi(s_0) = 3 + 0.3 \cdot 8.5 + 3.5 + 0.7 \cdot 6.5 = 6.5 + 2.55 + 4.55 = 13.6$

**Question 2.** Consider the following one-state MDP.




For a uniform random policy and your chosen $\gamma$ ensuring $0 < \gamma < 1$, calculate the value of the non-terminal state.

The policy given in the question is uniformly random:

$$\pi(L|0) = \pi(R|0) = \frac{1}{2}.$$

Also recall that the value of the terminal state is defined to be zero, i.e. $v_\pi(1) = 0$. Now using the Bellman equation for $v_\pi$, we can compute the value of the non-terminal state:

$$\begin{aligned} v_\pi(0) &= \pi(L|0) \cdot \Big[ -1 + \gamma v_\pi(0) \Big] + \pi(R|0) \cdot \Big[ +1 + \gamma v_\pi(1) \Big] \\ &= \frac{1}{2} \cdot \Big[ -1 + \gamma v_\pi(0) \Big] + \frac{1}{2} \cdot \Big[ 1 + \gamma \cdot 0 \Big] \\ &= \frac{1}{2} \gamma v_\pi(0). \end{aligned}$$

Since, $0 < \gamma < 1$, we can solve the above equation to obtain

$$v_\pi(0) = 0.$$

(Note that the value of the non-terminal policy is zero irrespective of what value we choose for the discount factor $\gamma$.)

**Question 3.** Exercise 3.11 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 4.** Exercise 3.12 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 5.** Exercise 3.13 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 6.** Exercise 3.14 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 7.** Exercise 3.15 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 8.** Exercise 3.17 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 9.** Exercise 3.18 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 10.** Exercise 3.19 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 11.** Exercise 3.22 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 12.** Exercise 3.25 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 13.** Exercise 3.26 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 14.** Exercise 3.27 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 15.** Exercise 3.28 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 16.** Exercise 3.29 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.